Sahil Rai

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PROFESSIONAL SUMMARY

AI/ML engineer (integrated BCA + MCA, 2027) building end-to-end Generative AI and RAG systems. Boosted document retrieval 35% in a 4-month internship, reached 0.844 ROC-AUC on video anomaly detection, and published a liver tumor characterization paper in Procedia Computer Science. Seeking entry-level AI/ML and LLM engineer roles.

INTERNSHIP

Data Science Intern – Navsoft, Mumbai, India | Apr 2026 – Jul 2026
Repository: github.com/SahilRai02/Company-website-chatbot-using-RAG-pipeline-and-LLM-intelligence

- **Application:** Engineered an enterprise AI assistant on a locally hosted LLM (Llama 3.1 via Ollama) with an async FastAPI backend and an interactive Streamlit front end.
- **RAG pipeline:** Built a multi-document pipeline with automated ingestion, chunking and semantic indexing using ChromaDB and Stella embeddings.
- **Hybrid retrieval:** Improved context retrieval accuracy by 35% by fusing vector search and BM25 with Reciprocal Rank Fusion (RRF) and adding cross-encoder re-ranking.
- **Reliability:** Reduced LLM hallucinations through prompt engineering, retrieval grounding and source citations on answers.
- **Backend:** Implemented asynchronous (async/await) FastAPI REST endpoints for non-blocking request handling, in a Docker-ready environment.

PROJECTS

Explain My Codebase | Python, RAG, Gemini, ChromaDB, LangChain

- Developed an AI developer tool that converts a Python codebase into 3 deliverables: an architecture diagram, a high-level summary and a detailed code walkthrough.
- Exported results as structured Markdown reports with embedded Mermaid architecture diagrams.
- Designed a 5-stage pipeline: AST analysis, code chunking, Gemini embeddings, ChromaDB similarity retrieval and Gemini LLM generation.

Weakly-Supervised Video Anomaly Detection | Python, PyTorch, OpenCV, R3D-18, BiGRU, MIL
Repository: github.com/SahilRai02/Weakly-supervised-anomoly-detection

- Built a surveillance anomaly detector on UCF-Crime (1,900 videos, 13 anomaly classes) using only video-level labels, with zero frame-level annotation.
- Extracted spatiotemporal features with a pretrained R3D-18, then modeled temporal patterns with a BiGRU trained using Multiple-Instance Learning (MIL).
- Achieved 0.844 ROC-AUC on 290 unseen test videos after training on 1,610 videos.

RESEARCH PUBLICATION

"Dual-Stage Computational Approach for Liver Tumor Characterization: U-Net 2D Spatial Analysis and XGBoost-Based Classification"

Procedia Computer Science, Vol. 282, Issue C, 2026, pp. 952–966 | DOI: doi.org/10.1016/j.procs.2026.05.138

- Proposed a dual-stage framework: a 2D U-Net for liver and tumor segmentation, followed by an XGBoost classifier for tumor characterization.
- Evaluated the framework on the public LiTS2017 benchmark dataset, using cross-validation to test robustness.

TECHNICAL SKILLS

GenAI & LLM: RAG, LLMs, Llama 3.1, Ollama, LangChain, Gemini API, Embeddings, Vector Databases, Semantic Search, Hybrid Search (BM25 + Vector, RRF), Cross-Encoder Re-ranking, Prompt Engineering

ML & Deep Learning: PyTorch, Scikit-Learn, XGBoost, OpenCV, Deep Learning, Computer Vision, CNNs, GRU, Multiple-Instance Learning, Model Evaluation (ROC-AUC)

Backend & Data: Python, FastAPI (Async), REST APIs, Node.js (Basic), Streamlit, MySQL, ChromaDB, Sentence Transformers

Tools: Docker, Git, GitHub, Linux, MLflow, AWS (Cloud Fundamentals)

CERTIFICATIONS

- **AWS Cloud Practitioner Essentials** (Completion Certificate) – AWS Training & Certification, Oct 2026
- **Node (Basic)** (Skill Certification, passed) – HackerRank, Oct 2026 | Credential ID: 498CA97A9441

EDUCATION

Amity University, Noida, India – BCA + MCA Integrated Program | 2022 – Expected 2027
CGPA: 7.87 / 10.0 | Coursework: Data Structures & Algorithms, Machine Learning, Deep Learning, DBMS, Software Engineering

Shree Swaminarayan Vidhyalay, Surat, Gujarat – Class XII (80%) | 2021 – 2022